Progress report - Winfinity

Jharkhand Societal Innovation Collaboration Portal

1. Name of the problem statement

AI-Powered Societal Innovation Collaboration Portal Bridging Citizen Challenges, Higher Education Institutions, and Industry CSR

2. An abstract of the proposed idea

The Jharkhand Societal Innovation Collaboration Portal is an AI-powered closed-loop platform designed to transform grassroots societal challenges across all 24 districts into actionable university research and corporate-backed deployments, fulfilling the experiential learning mandates of the National Education Policy (NEP 2020). Citizens and district authorities submit hyper-local community problems in domains such as water purification, agriculture, healthcare, rural livelihoods, and sanitation through a guided four-step intake wizard with geolocated photo evidence. An integrated dual-mode intelligence pipeline powered by Google Gemini 1.5 Flash and deterministic NLP performs instant problem categorization, priority scoring, duplicate detection, and academic discipline extraction. A multi-criteria matching algorithm dynamically recommends the top three Higher Education Institutions (HEIs) in Jharkhand based on departmental strengths and regional proximity. Matched university faculty and students establish multidisciplinary project workspaces to formulate stage-gated technical proposals and track milestone progress. Simultaneously, corporate enterprises access a transparent Corporate Social Responsibility (CSR) marketplace to pledge capital, donate equipment, and sponsor vetted initiatives. The portal closes the innovation loop through verified citizen feedback ratings and real-time state telemetry tracking ground-level socio-economic impact across Jharkhand.

System Architecture and Stakeholder Workflow:

Stage	Functional Description
1. Citizen Intake	4-step progressive wizard with automatic GPS geocoding and photo evidence uploads to cloud storage across all 24 districts.
2. AI Problem Triage	Dual-mode pipeline executing Google Gemini 1.5 Flash structured inference with a deterministic keyword NLP fallback for zero-downtime offline operation.
3. HEI Auto-Matching	Explainable 40/30/20/10 weighted formula matching top state institutions (BIT Mesra, IIT ISM Dhanbad, BIT Sindri, BAU, CUJ, RIMS) based on domain specialization and proximity.
4. Project & CSR Hub	Multidisciplinary student-faculty rosters, stage-gated proposals, milestone tracking, corporate CSR capital pledging, and ground impact telemetry.

3. Work done/implemented till 10PM today i.e. 08/09/2026

Engineered end-to-end full-stack portal featuring 4-step citizen problem intake with photo uploads, dual-mode Gemini/NLP AI triage, 40/30/20/10 HEI matching algorithm, multidisciplinary university project workspaces with milestone tracking, corporate CSR sponsorship pledges, 5-role RBAC authentication, 24-district interactive command center map, and verified impact analytics.

Implemented Subsystems and Technical Components:

Subsystem	Implementation Details	Status
Citizen Intake Wizard	4-step guided intake form with GPS coordinate tagging, photo preview, and Supabase Storage bucket uploads.	Completed
Dual AI Diagnostics	Google Gemini 1.5 Flash structured parser with deterministic keyword NLP fallback for category and priority calculation.	Completed
HEI Matching Engine	40/30/20/10 weighted formula ranking top 3 institutions based on domain, discipline overlap, capability, and location.	Completed
University Workspaces	Cross-department team builder, 7-stage project proposals, milestone tracking checklists, and submission workflows.	Completed
CSR Marketplace	Corporate sponsorship directory with express-interest pledge modal for financial capital, equipment, and mentorship.	Completed
State Command Center	Interactive 24-district heatmap visualizer and 8 visual analytics charts covering state grievance metrics.	Completed

4. Work left

Integrating real-time SMS/WhatsApp multilingual citizen notifications in Hindi and Santhali, embedding automated blockchain-based certificate validation for academic credits, onboarding live payment gateways for direct CSR grant disbursements, and conducting field stress-testing with regional district collectors and university pilot teams across Santhal Pargana and Kolhan divisions.

Upcoming Milestone Targets:

Deliverable	Technical Scope	Target Date
SMS & WhatsApp Gateway	Automated citizen status updates in Hindi, Santhali, and English via messaging APIs.	09/09/2026
Academic Credit Attestation	Verifiable credentialing for student experiential learning credits under NEP 2020.	09/09/2026
Direct CSR Payment Gateway	Escrow disbursement integration linked directly to milestone verifications.	09/09/2026
Regional Pilot Testing	Field stress-testing with pilot institutions (BIT Sindri, BAU) and district collectors.	09/09/2026

5. Challenges faced

Challenge 1: Zero-downtime dual AI intelligence and offline resilience for rural connectivity

In remote tribal regions of Jharkhand (such as Latehar, Pakur, and Khunti), low mobile data bandwidth and intermittent network connectivity frequently cause external cloud LLM API calls to timeout, which previously risked stalling citizen problem intake and urgency prioritization during field surveys. This was resolved by architecting a dual-mode data layer: while Google Gemini 1.5 Flash provides deep multidimensional inference when online, an instantaneous deterministic rule-based NLP classifier and heuristic priority engine activates automatically during network drops, guaranteeing continuous offline operation and zero data loss.

Challenge 2: Explainable multi-institutional matching balancing academic rigor and regional proximity

Premier central institutions possess extensive department catalogs, creating an algorithmic bias where standard similarity algorithms continuously assigned them every challenge, thereby marginalizing specialized regional state universities such as Birsa Agricultural University and BIT Sindri. This was resolved by designing an explainable 40/30/20/10 weighted matching formula incorporating domain specialization (40%), discipline overlap (30%), institutional capability (20%), and district geographic proximity (10%), mathematically balancing institutional prestige with regional domain authority to ensure equitable project allocation across all 24 districts.

Submitted by Team Winfinity
Jharkhand Societal Innovation Collaboration Portal